

NMAM INSTITUTE OF TECHNOLOGY, NITTE
(An Autonomous Institution affiliated to VTU, Belagavi)
Sixth Semester B.E. (ISE) (Credit System) Degree Examinations
 May - June 2023

20IS602 / 17IS602 - COMPUTER NETWORKS

Duration: 3 Hours

Max. Marks: 100

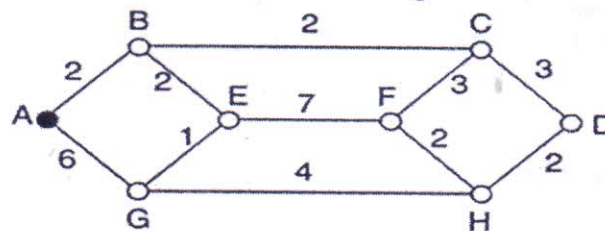
*Note: Answer **Five full** questions choosing **One full** question from **each Unit**.*

Unit – I

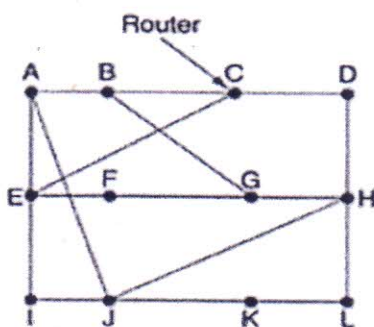
- | | Marks | BT* | CO* | PO* |
|--|-------|-----|-----|-----|
| 1. a) With a neat diagram, explain IEEE 802 LAN standards. Also explain MAC and LLC. | 10 | L*2 | 1 | 2 |
| b) Explain Fast Ethernet and Giga bit Ethernet. | 10 | L3 | 1 | 2 |
| 2. a) Write a note on i) BSS ii) ESS iii) Ad hoc network iv) AP. | 10 | L3 | 1 | 2 |
| b) Write the spanning tree algorithm. Explain the same with an example. | 10 | L3 | 1 | 2 |

Unit – II

3. a) Consider the subnet diagram shown below. Find the minimum distance between A and using shortest path routing algorithm. Show all the intermediate steps in the diagram.



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|---|----|----|---|---|
| b) Distinguish between Datagram and Virtual Circuit subnets with respect to the design issues. | 10 | L3 | 2 | 2 |
| c) Explain advantages and disadvantages of flooding and list any two ways to avoid packet flooding in the network. | 6 | L3 | 2 | 2 |
| | 4 | L2 | 2 | 2 |
| 4. a) Explain hierarchical routing with an example. | 10 | L3 | 2 | 2 |
| b) Construct the new routing table for router J of a given subnet using the data received from the neighbour routers. | | | | |



To	A	I	H	K
A	0	24	20	21
B	12	36	31	28
C	25	18	19	36
D	40	27	8	24
E	14	7	30	22
F	23	20	19	40
G	18	31	6	31
H	17	20	0	19
I	21	0	14	22
J	9	11	7	10
K	24	22	22	0
L	29	33	9	9

JA delay is 8 JI delay is 10 JH delay is 12 JK delay is 6

Vectors received from J's four neighbors

10 L3 2 2

P.T.O.

Unit – III

- | | | | | | |
|-------|---|----|----|---|---|
| 5. a) | Write IPV4 header format and explain the significance of each field. | 10 | L2 | 3 | 2 |
| b) | Explain the various ways to control congestion in datagram subnet. | 10 | L2 | 3 | 2 |
| 6. a) | Differentiate between IPV4 and IPV6 header format. | 10 | L2 | 3 | 2 |
| b) | Explain expedited forwarding and assured forwarding rules in fragmentation. | 6 | L2 | 3 | 2 |
| c) | List and explain any 4 techniques for achieving good quality of Service. | 4 | L2 | 3 | 2 |

Unit – IV

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|-------|--|----|----|---|---|
| 7. a) | With an activity diagram, explain TCP Three-Way handshake to establish the connection between two stations. | 10 | L2 | 4 | 2 |
| b) | Explain the pros and cons of RARP, BOOTP and DHCP protocols. | 6 | L2 | 4 | 2 |
| c) | Differentiate between flow control and congestion control in TCP. | 4 | L2 | 4 | 2 |
| 8. a) | Explain the significance of connection termination in TCP and also show the activity diagram to the process. | 8 | L2 | 4 | 2 |
| b) | List any 6 messages generated by the ICMP protocol and brief the significance of each message type. | 6 | L2 | 4 | 2 |
| c) | Differentiate chained fixed size and variable size buffer with a neat diagram. | 6 | L2 | 4 | 2 |

Unit – V

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|--------|--|----|----|---|---|
| 9. a) | Explain the working of Real time transport protocol (RTP). | 10 | L3 | 5 | 2 |
| b) | Explain TCP congestion control algorithm by showing the phases of congestion using a neat diagram. | 10 | L3 | 5 | 2 |
| 10. a) | Draw the TCP header format and brief the usage of each field in the header. | 10 | L3 | 5 | 2 |
| b) | List and explain the various problems that can degrade the performance of TCP. | 10 | L3 | 5 | 3 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
